

Bidirectional Spoken-Written Text Conversion with Large Language Models

Muyeol Choi^{1*}, HyunJung Choi^{2*}, Yohan Lim², Jeonguk Bang¹, Minkyu Lee¹,
Seonhui Kim², Seung Yun¹, Donghyun Kim¹, Minsoo Kim¹, SangHun Kim^{1†}

¹Electronics and Telecommunications Research Institute, South Korea

²University of Science and Technology, South Korea

mychoi@etri.re.kr*, chocohj@etri.re.kr*, ksh@etri.re.kr†

Abstract

Traditional ASR systems normalize transcriptions into spoken form for training, leading to spoken form outputs. In contrast, modern Transformer-based models directly map speech-to-text, preserving the written form. However, existing speech databases mainly contain spoken form, causing inconsistencies in recognition results. To address this, Inverse Text Normalization (ITN) is required, and dual transcription data is crucial for effective training. However, constructing such datasets is costly and time-consuming. This study proposes a data augmentation method leveraging LLMs to automatically generate dual transcription data with minimal effort. It employs iterative learning to expand the dataset through supervised and semi-supervised methods. Additionally, the bidirectional text conversion (BTC) model supports both ITN and TN within a unified framework. Experimental results demonstrate that the proposed method achieved an ERR of 13.4% and 4.7%, outperforming prior approaches.

Index Terms: Inverse Text Normalization, Spoken-to-Written, Written-to-Spoken, Bidirectional Text Conversion

1. Introduction

In the era of artificial intelligence, speech recognition technology has advanced beyond simple speech-to-text conversion, now requiring context-aware accuracy and well-formatted output. Depending on context, the phrase ‘Twenty twenty-five’ may refer to a year (2025), a time (20:25), or a score (20–25), demonstrating how its written form varies depending on meaning. Similarly, long numerical expressions are typically represented in digit form (e.g., 2,345,600) rather than full-text transcriptions, enhancing readability and comprehension. Many other numerical and symbolic expressions require precise formatting to ensure clarity.

Conventional speech recognition systems traditionally are based on spoken-form transcriptions for training, leading to outputs that reflect pronunciation rather than written form. This leads to inconsistencies when written form output is required, making Inverse Text Normalization (ITN) [1, 2, 3, 4] has become essential for converting spoken form transcriptions back into written form. Recently, Transformer-based end-to-end speech recognition models [5, 6, 7] have gained prominence for directly converting speech signals into text. These models have demonstrated strong capabilities in learning mappings between speech signals and written form, enabling more effective handling of numbers, symbols, and foreign words. However, they rely on large-scale databases for training,

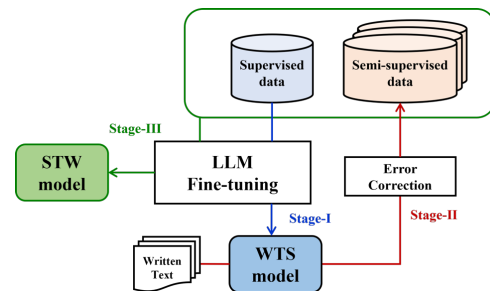


Figure 1. Iterative data augmentation method based on semi-supervised learning

making manual transcription efforts infeasible. As an alternative, constructing large-scale speech databases from broadcast data reduces manual effort but often introduces transcription errors and inconsistencies, degrading recognition performance [8]. As a result, speech databases can contain a mix of spoken form and written form transcriptions, leading to further inconsistencies in recognition outputs.

To resolve this issue, Transformer-based ITN techniques have been proposed [9]. Inspired by the success of Transformer models in machine translation, this approach frames spoken-to-written form conversion as a translation task. Just as parallel data pairs are essential for training effective machine translation models, dual transcription data—where both spoken form and written form transcriptions are recorded simultaneously—is critical for ITN model training. Building on this approach, an LLM-based ITN technique has been proposed [10]. Fine-tuning an LLM on dual transcription data has shown improved contextual awareness, achieving superior ITN performance compared to conventional Transformer-based approaches. Despite these advancements, Korean dual transcription data remains scarce, posing a major challenge to ITN development [11, 12]. Constructing such datasets is costly and time-consuming due to its heavy reliance on manual work, making large-scale data construction impractical.

To address this issue, we introduce a data augmentation approach leveraging LLMs to automatically generate dual transcription data, as shown in Figure 1. The proposed method follows an iterative data augmentation strategy, progressively expanding the dataset through supervised and semi-supervised learning. As part of this approach, we propose a Written-to-Spoken (WTS) conversion method, which applies LLM-based Text Normalization (TN) to transform written form into spoken form. This method automates the construction of the dataset, significantly reducing manual effort. Additionally, an error

* These authors contributed equally to this work.

- (a) Dual transcription structure: (written form)/(spoken form)
 (b) Dual transcription example: 소음은 최대 (이십)/(20)(데시벨)/(dB) 작다. (*The noise level is up to (20)/(twenty)(decibels)/(dB) lower.*)
 (c) Data format for Finetuning
 :<SPOKEN>[BOS] 평균 영 점 사마이크로미터 크기의 미세먼지를 구십사퍼센트 이상 걸러내는 제품이다.[EOS]
 <WRITTEN>[BOS] 평균 0.4 μm 크기의 미세먼지를 94% 이상 걸러내는 제품이다.[EOS]
 (<SPOKEN>[BOS] It is a product that filters more than ninety-four percent of fine particles with an average size of zero point four micrometers.[EOS]<WRITTEN>[BOS] It is a product that filters more than 94% of fine particles with an average size of 0.4 μm .[EOS])

Figure 2. Examples of dual transcription and data format (The part written in italics is the English translation of the Korean sentence.)

detection and correction process enhance data quality. Furthermore, we introduce the Bidirectional Text Conversion (BTC) model, which integrates both ITN and TN conversions, enabling flexible text transformations across different domains.

The structure of this paper is as follows: Section 2 describes the LLM-based ITN model. Section 3 details the proposed method. Section 4 presents and analyzes the experimental results. Finally, Section 5 provides the conclusion.

2. Baseline Model for ITN

2.1. Korean LLM Trained from Scratch (GPT-2-Kor)

The baseline model for ITN is built upon GPT-2-Kor, a lightweight Korean LLM trained from scratch using a Korean corpus. GPT-2-Kor follows the GPT-2 [13] small architecture and is optimized for computational efficiency.

2.2. Korean ITN Model (Spoken-to-Written, STW)

GPT-2-Kor was fine-tuned as an STW model for Korean ITN. To train the model, dual transcription data was reformatted to capture transformation patterns between spoken form and written form. The structure of the dual transcription data and the training set used in the STW model are presented in Figure 2. Figure 2(a) presents the dual transcription format, aligning spoken form and written form transcriptions in parallel. When spoken and written forms differ within a sentence, the transcription follows the parallel format shown in Figure 2(b). Figure 2(c) demonstrates the structured input format used for model training:

“<SPOKEN> [BOS] spoken form sentences [EOS]
 <WRITTEN> [BOS] written form sentences [EOS]”

Here, ‘<SPOKEN>’ and ‘<WRITTEN>’ are special tokens indicating the transcription type, while [BOS] (Begin of Sentence) and [EOS] (End of Sentence) mark sentence boundaries.

3. Iterative Data Augmentation Based on Semi-Supervised Learning

A major challenge in improving STW model performance is the difficulty of obtaining large-scale training data, as dual transcription datasets are primarily constructed manually. To address this issue, this study proposes a novel approach, introducing a new LLM-based Text Normalization (TN) technique for Written-to-Spoken (WTS)—the reverse process of STW.

WTS generates spoken form outputs from written form inputs, enabling automated construction of dual transcription datasets and significantly reducing manual effort. The overall framework for data generation and iterative training using both STW and WTS models is illustrated in Figure 1.

3.1. Supervised Data-Based WTS Model Training (Stage-I)

The proposed method follows an iterative data augmentation strategy, progressively expanding dual transcription data through supervised and semi-supervised learning. In Stage-I, supervised data is used to fine-tune GPT-2-Kor into the WTS model. The model is trained by modifying the token structure of the STW training dataset as follows:

“<WRITTEN> [BOS] written form sentences [EOS]
 <SPOKEN> [BOS] spoken form sentences [EOS]”

By learning this mapping, the model generates spoken form outputs from written form inputs, forming the basis for semi-supervised learning in the next stage.

3.2. Semi-Supervised Data Generation Using the WTS Model (Stage-II)

In Stage-II, the WTS model is used to automatically generate spoken form outputs, expanding the dataset beyond the manually constructed samples. To ensure data quality, a conversion error detection and correction process is applied. The primary sources of errors stem from numbers, symbols, and foreign words. Since manually refining all error patterns is inefficient, the correction process focuses on symbol transformations. A list of frequently occurring erroneous characters was compiled, followed by automatic validation and correction. The verified data was incorporated into the existing supervised dataset, continuously improving the model through iterative learning.

Table 1 presents a comparison of the input text, WTS model output, and the final dual transcription data after error correction. For example, the incorrect transformation of ‘ha’ to ‘에이커(acre)’ was corrected to ‘헥타르(hectare)’ before finalizing the transcription. The corrected words are integrated into the dual transcription format, ensuring that the final data follows the structured example shown in the last row of Table 1. Stage-II enables the system to expand and refine large-scale dual transcription datasets with minimal human intervention, significantly enhancing data availability and quality.

Table 1. Example of dual-transcription generated by the WTS model

Input Text	오염된 지역의 범위는 250 ha에 이른다. (<i>The contaminated area's range reaches 250 ha.</i>)
WTS Output	오염된 지역의 범위는 이백오십 에이커에 이른다. (<i>The contaminated area's range reaches two hundred fifty acres.</i>)
Error Correction	오염된 지역의 범위는 이백오십 헥타르에 이른다. (<i>The contaminated area's range reaches two hundred fifty hectares.</i>)
Generated Dual Trans.	오염된 지역의 범위는(이백오십)/(250)(헥타르)/(ha)에 이른다. (<i>The contaminated area's range reaches (two hundred fifty)/(250) (hectares)/(ha).</i>)

3.3. STW Model Update Using Augmented Dataset (Stage-III)

With the augmented dataset incorporating both supervised and semi-supervised data, the STW model undergoes iterative fine-tuning in Stage-III. The augmented dataset, now including verified spoken form outputs, is used to retrain the STW model. Through this iterative learning process, the STW model benefits from an enhanced diversity of training examples, leading to improved accuracy and adaptability. The final STW model is expected to handle a wider range of linguistic variations and domain-specific ITN more effectively.

3.4. Bidirectional Text Conversion (BTC)

In certain domains, such as air traffic control and military communication, spoken form is preferred over numerical values to prevent misinterpretation. Since different domains may require distinct transcription styles, speech recognition outputs do not always need to be in written form. To address this, we designed the Bidirectional Text Conversion (BTC) model, which integrates both STW and WTS conversions into a single model, allowing for flexible text transformations based on domain-specific requirements.

The BTC model is fine-tuned using GPT-2-Kor and the augmented dataset. It utilizes the data generated in Stage-III, incorporating both STW and WTS training formats simultaneously, which provides a simple yet effective approach for bidirectional text conversion. Figure 3 illustrates the inference process of the BTC model, which performs bidirectional text conversion by taking spoken form (S) as input and generating written form (W) as output, or vice versa, depending on the inference direction. Prompts can also selectively apply ITN or TN based on the intended application.

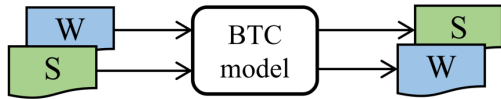


Figure 3. The inference process of the BTC model

4. Experiments and Results

The Korean LLM model was trained on corpora provided by AIHub [14]. The dataset consists of commercially available book-based Korean corpora, specialized medical and legal texts, National Assembly transcripts, and newspaper articles collected from Modu Corpus [15], as well as sentences from Korean Wikipedia and Namuwiki, amounting to 7.6 billion (B) tokens for training. For fine-tuning, lecture content data from AIHub [16] and news article data from ETRI [17] were used, resulting in 4.5 million (M) fine-tuned sentences. Performance evaluation focused on text that captures spoken-to-written form variations, particularly in numbers, symbols, and foreign words. These categories are collectively referred to as target words, as they are key to assessing the model's ability to convert spoken form into written form. The evaluation dataset consists of Set A and Set B:

- Set A: Contains 3,000 news texts from 2002–2003, created during the initial dual transcription data construction.
- Set B: Contains 3,000 news texts from 2022.

The evaluation was conducted by measuring Inverse Character Error Rate (I-CER) and Non-Inverse Character Error Rate (NI-CER). I-CER assesses the model's accuracy in converting target characters, while NI-CER evaluates its ability to preserve non-target characters in their original form. The analysis of target word proportions (numerals and alphabetic characters) in the evaluation datasets is presented in Table 2. Set B contains a higher proportion of target words compared to Set A, but the ratio of numerals and alphabetic characters remains similar across both datasets.

Table 2. Proportion (%) of numerals and alphabetic characters in the evaluation set

	Proportion of Target Words	Numerals in Target Words	Alphabets in Target Words
Set-A	8.0	88.4	11.6
Set-B	15.0	86.6	13.4

Table 3 presents the performance comparison of WTS models fine-tuned on two different Korean LLMs using 4.5M supervised learning sentences. One is GPT-2-Kor (G2K, 112M parameters) [18], and the other is Llama-3-Open-Kor-8B (L3K, 8B parameters). The Llama-3-Open-Kor-8B model is a multilingual variant of Llama-3 [19, 20], further trained on Korean data.

Table 3. Performance comparison of WTS models with different LLMs

LLMs	I-CER(%)			NI-CER(%)		
	Set-A	Set-B	Total	Set-A	Set-B	Total
G2K	3.46	8.27	6.70	0.04	0.22	0.12
L3K	3.22	9.40	7.37	0.01	0.03	0.02

Experimental results indicate that model size alone does not determine WTS performance, as the smaller GPT-2-Kor outperformed the much larger Llama-3-Open-Ko-8B. The WTS model generated 7.5M spoken form sentences, bringing the total training data to 12M when combined with 4.5M supervised sentences. While additional data can be continuously generated, this study focuses on an initial augmentation phase.

Table 4 shows the performance of the proposed STW model and the baseline model. Both models were fine-tuned using GPT-2-Kor, but with different amounts of training data. The baseline model used 4.5M supervised sentences, while the proposed model was trained with 12M sentences.

Table 4. Performance comparison of STW models

	I-CER(%)			NI-CER(%)		
	Set-A	Set-B	Total	Set-A	Set-B	Total
Baseline[10]	5.83*	11.16	9.35	0.20*	0.44	0.31
Proposed	8.42	2.37	4.41	0.31	0.06	0.20

* The baseline model's performance was re-evaluated due to a bug in the metric calculation in [10], leading to updated results with improved accuracy.

The proposed model performed slightly worse than the baseline model on Set A, likely due to differences in number notation styles. Since Set A was constructed based on past news text, it inevitably reflects the numerical notation conventions of that period. Set A tends to represent numerical units in Korean

Table 6. Examples of ITN results for the baseline and proposed method
(The part written in *italics* is the English translation of the Korean sentence.)

Input	이와삼분의이닝 동안 삼진 세 개를 잡아내며 상대 타선을 완벽히 봉쇄했다. (He dominated the opposing lineup, striking out <i>three over two and two-thirds</i> innings.)
Baseline	2와3분의2이닝 동안 삼진 3개를 잡아내며 상대 타선을 완벽히 봉쇄했다. (He dominated the opposing lineup, striking out 3 over <i>2 and 2/3</i> innings.)
Proposed	2 ² / ₃ 이닝 동안 삼진 3개를 잡아내며 상대 타선을 완벽히 봉쇄했다. (He dominated the opposing lineup, striking out 3 over <i>2²/₃</i> innings.)
Input	초미세 먼지 농도는 세제곱미터당 이십삼마이크로그램 으로 '보통' 수준이었다. (The fine dust concentration was twenty-three <i>micrograms per cubic meter</i> , which is considered 'moderate'.)
Baseline	초미세 먼지 농도는 m3 당 23ug 으로 '보통' 수준이었다. (The fine dust concentration was 23ug per <i>m3</i> , which is considered 'moderate'.)
Proposed	초미세 먼지 농도는 m ³ 당 23 μg 으로 '보통' 수준이었다. (The fine dust concentration was 23μg per <i>m³</i> , which is considered 'moderate'.)

words (e.g., ‘백’(hundred), ‘천’(thousand), ‘만’(ten thousand)) instead of Arabic numerals, leading to mismatches with reference answers despite identical meanings. For example, ‘200’ and ‘2 백’(two hundred) are semantically equivalent but differ in notation, which may impact performance. To ensure automated evaluation and consistent scoring, the reference answers were fixed rather than dynamically adjusted, further limiting the flexibility in handling variations in notation. However, for Set B, which contains more complex expressions and follows a recently standardized notation system, the proposed model achieved I-CER and NI-CER of 2.37% and 0.06%, respectively, significantly outperforming the baseline model, with an Error Reduction Rate (ERR) of 52.8% for I-CER and 35.4% for NI-CER.

Table 5 compares the proposed BTC model against independent STW and WTS models. The independent models were fine-tuned on 12M sentences, while the BTC model leveraged both directions of the dataset, effectively doubling the training data. However, as the data comes from existing dual transcription datasets, no additional construction cost is incurred. Additionally, the BTC model integrates both ITN and TN within a single framework. For the STW task, the BTC model achieved an ERR of 13.4% for I-CER and 25.0% for NI-CER. These values are derived from the total score, where BTC improved the independent STW model’s I-CER from 4.41% to 3.82% and NI-CER from 0.20% to 0.15%. For the WTS task, the BTC model achieved an ERR of 4.7% in I-CER and 50.0% in NI-CER, reducing the independent WTS model’s I-CER from 4.19% to 3.99% and NI-CER from 0.22% to 0.11%. These

Table 5. Performance comparison of BTC model with independent of STW, WTS models

		I-CER(%)		
	Task	Set-A	Set-B	Total
Indep.	STW	8.42	2.37	4.41
	WTS	3.43	4.56	4.19
BTC	STW	6.12	2.64	3.82
	WTS	3.09	4.43	3.99
		NI-CER(%)		
	Task	Set-A	Set-B	Total
Indep.	STW	0.31	0.06	0.20
	WTS	0.38	0.02	0.22
BTC	STW	0.22	0.07	0.15
	WTS	0.17	0.02	0.11

results demonstrate that the BTC model consistently outperforms independent models in both ITN and TN tasks. Furthermore, by leveraging existing data, the BTC model achieves these improvements without additional data collection costs.

Table 6 presents a qualitative comparison of outputs generated by the baseline model and the proposed STW model. In the first example, the spoken form ‘이와삼분의이닝’ corresponds to ‘three over two and two-thirds innings.’ The baseline model rendered the fraction in a mixed format combining numbers and words (2 and 2/3), whereas the proposed STW model produced a more concise and mathematically precise representation (2²/₃), improving readability and clarity. In the second example, ‘세제곱미터’ (cubic meter) and ‘마이크로그램’ (microgram) were converted differently. The baseline model generated ‘m3’ and ‘ug’, using a combination of numerals and alphabetic characters instead of their standard scientific symbols. While ‘m³’ is normalized to ‘m3’ during evaluation, resulting in no quantitative difference, ‘m3’ is not the optimal notation. In contrast, the proposed model successfully rendered these units in their standard scientific notation (m³ and μg), demonstrating improved handling of numerical and scientific symbols. These results highlight the proposed STW model’s enhanced ability to process numerical values and scientific symbols, enabling more precise and structured conversions.

5. Conclusions

This study investigated ITN techniques to address inconsistencies between spoken form and written form transcription in speech recognition outputs. To tackle this issue, we proposed an iterative data augmentation method based on semi-supervised learning, leveraging LLMs. The proposed approach automatically generates dual transcription data, reducing costs while meeting ITN requirements across various domains. Additionally, we implemented a bidirectional transformation model, enabling a single model to perform both ITN and TN. Experimental results showed that the proposed method achieved an ERR of 13.4% and 4.7% compared to independent models, demonstrating superior performance and efficiency over existing approaches.

For future research, we plan to explore prompt-based techniques to develop domain-optimized transformation models, allowing the system to adapt to diverse domains and complex contexts.

6. Acknowledgements

This study was supported by an Electronics and Telecommunications Research Institute (ETRI) grant funded by the Korean Government (25ZB1100, Core Technology Research for Self-improving Integrated Artificial Intelligence Systems).

7. References

- [1] Ihori, M., et al. "Transcribing speech as spoken and written dual text using an autoregressive model." In *Proceedings of Interspeech 2023*, 2023.
- [2] Antonova, A., et al. "Thutmose Tagger: Single-pass neural model for Inverse Text Normalization." In *Proceedings of Interspeech 2022*, 2022.
- [3] Phan, T. A., et al. "Neural inverse text normalization with numerical recognition for low resource scenarios." In *Proceedings of ACIIDS 2022, Lecture Notes in Computer Science*, vol. 13757, pp. 582–594, 2022.
- [4] Khare, Y., et al. "Universal-2-TF: Robust all-neural text formatting for ASR." *arXiv preprint arXiv:2501.05948*, 2025.
- [5] Vaswani, A., et al. "Attention is all you need." In *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [6] Karita, S., et al. "Improving transformer-based end-to-end speech recognition with connectionist temporal classification and language model integration." In *Proceedings of Interspeech 2019*, 2019.
- [7] Moritz, N., Hori, T., and Le, J. "Streaming automatic speech recognition with the transformer model." In *Proceedings of ICASSP 2020*, 2020.
- [8] Bang, J.-U., et al. "Automatic construction of a large-scale speech recognition database using multi-genre broadcast data with inaccurate subtitle timestamps." *IEICE Transactions on Information and Systems*, vol. 103, no. 2, pp. 406–415, 2020.
- [9] Choi, H., et al. "Spoken-to-written text conversion for enhancement of Korean–English readability and machine translation." *ETRI Journal*, vol. 46, no. 1, pp. 127–136, 2024.
- [10] Choi, H., et al. "Spoken-to-written text conversion with Large Language Model." In *Proceedings of Interspeech 2024*, 2024.
- [11] Paul, D., et al. "Improving data-driven inverse text normalization using data augmentation and machine translation." In *Proceedings of Interspeech 2022*, 2022.
- [12] Chen, S.-J., et al. "Language agnostic data-driven inverse text normalization." In *Proceedings of Interspeech 2023*, 2023.
- [13] Radford, A., et al. "Language models are unsupervised multitask learners." *OpenAI Blog*, vol. 1, no. 8, pp. 9, 2019.
- [14] AIHub. [Online]. Available: <https://www.aihub.or.kr/>.
- [15] National Institute of Korean Language. "Korpora." Available: <https://kli.korean.go.kr/corpus/main/requestMain.do>.
- [16] AIHub. "AIHub Korean lecture speech dataset." 2020. [Online]. Available: <https://www.aihub.or.kr/>.
- [17] ETRI. "ETRI Korean common speech dataset." 2004. [Online]. Available: <https://www.etri.re.kr/eng/main/main.etri>.
- [18] Lee, J. "Llama-3-Open-Ko." Hugging Face, 2024. [Online]. Available: <https://huggingface.co/beomi/Llama-3-Open-Ko-8B>.
- [19] Dubey, A., et al. "The Llama 3 herd of models." *arXiv preprint arXiv:2407.21783*, 2024.
- [20] Meta AI. "Introducing Meta Llama 3: The most capable openly available LLM to date." 2024. [Online]. Available: <https://ai.meta.com/blog/meta-llama-3/>.